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Temporal Representation Enhancement (TRE): Learning to Forget Dominant Patterns for Enhanced Temporal Spiking Features

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Abstract

Spiking Neural Networks (SNNs) naturally process visual inputs across multiple timesteps, offering rich temporal dynamics and energy-efficient computation. However, the temporally invariant supervision commonly used in training tends to reinforce the same dominant response patterns across timesteps, leading to redundant representations and limiting temporal discriminability. To overcome this limitation, we introduce Temporal Representation Enhancement *(TRE), a novel learning-to-forget paradigm that encourages more diverse and complementary temporal representations. TRE identifies high-contribution semantic patterns through class-specific contribution estimation and temporal accumulation, and selectively suppresses them using a dynamic modulation strategy. By redirecting the model's attention toward alternative yet informative semantic cues, TRE promotes the learning of complementary features across timesteps. This approach not only strengthens the temporal discriminative capacity of SNNs but also enables more effective multi-timestep learning by leveraging richer semantic information. Extensive experiments on both static image datasets and dynamic neuromorphic datasets demonstrate that TRE consistently improves classification accuracy and feature diversity across different SNN backbones.*

1. Introduction

Spiking Neural Networks (SNNs) have emerged as a promising class of biologically inspired models that bridge neuroscience and artificial intelligence [34, 50, 52, 64]. Unlike conventional Artificial Neural Networks (ANNs) that operate with continuous and static activations, SNNs communicate through discrete spike events, emulating the dynamic firing behavior of biological neurons [42, 65]. This event-driven mechanism allows SNNs to naturally capture temporal dependencies and sequential information, closely resembling the dynamic nature of human perception and cognition. Moreover, SNNs are inherently low-power, making them highly attractive for energy-efficient and neuromorphic applications [7, 18, 46, 48, 56, 61].

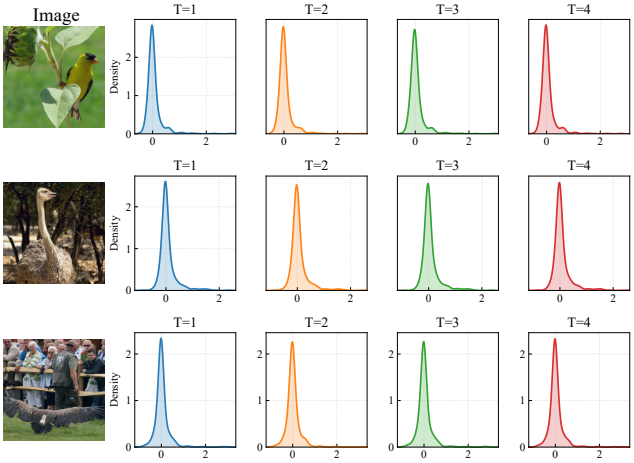


Figure 1. Channel contribution density distributions of SNNs across timesteps (T=1–4) on ImageNet samples. The narrow and highly peaked curves indicate strong activation dominance and low temporal diversity in conventional SNNs representations.

Despite substantial advances, effectively exploiting the rich temporal dynamics embedded in multi-timestep spiking activity remains a fundamental challenge. Most directly trained SNNs [54, 55, 62, 63] adopt surrogate-gradient backpropagation with identical supervision across timesteps. While this design stabilizes optimization, it enforces uniform gradient dynamics that cause features from different timesteps to converge toward similar representational subspaces. As a result, the network repeatedly ac-

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tivates overlapping high-response channels (Fig. 1), leading to narrow, sharply peaked channel-contribution distributions (T=1–4) that reveal strong activation dominance and limited temporal diversity—evidence that conventional SNNs struggle to exploit complementary temporal representations.

To enhance temporal representation learning, a series of direct-training strategies have been explored. Early work such as Spatio-Temporal Backpropagation (STBP) [58] exploited the leaky-integrate-and-fire (LIF) model to jointly utilize spatial and temporal dependencies, achieving strong results on shallow networks. Later studies improved stability and convergence in deeper SNNs through normalization methods analogous to batch normalization (BN) in ANNs, including threshold-dependent BN (tdBN) [68], temporal BN (BNTT) [32], postsynaptic-potential BN (PSP-BN) [30], and time-effective BN (TEBN) [12]. From an architectural perspective, SEW-ResNet [13] refined residual connections via spike-element-wise identity mapping, whereas neuron-level optimizations such as ReL-PSP [63] and PLIF [14] adjusted membrane dynamics to improve flexibility. From a gradient-optimization viewpoint, Temporal Efficient Training (TET) [11] leveraged temporal surrogate gradients to mitigate momentum loss, while Spatial Learning Through Time (SLTT) [45] showed that discarding temporal gradients can further boost efficiency. Although these efforts have advanced directly trained SNNs, they still rely on identical timestep supervision, which ultimately drives representations to collapse into similar subspaces and restricts temporal diversity. Meanwhile, masking-based visual recognition methods [27, 57] have shown that suppressing dominant spatial regions can reveal alternative cues, yet such techniques remain confined to static domains and fail to address the temporal redundancy inherent to spike-based learning.

To overcome these challenges, we propose the Temporal Representation Enhancement (TRE) framework, which reformulates temporal learning as a *learning-to-forget* process. Instead of repeatedly reinforcing dominant features across timesteps, TRE dynamically suppresses high-contribution semantic channels while promoting the discovery of complementary representations. Specifically, TRE employs a class-specific contribution estimation mechanism to measure each channel's relevance to the classification objective and applies a hybrid modulation strategy combining fixed and adaptive thresholds. This targeted suppression adaptively attenuates over-dominant features, guiding subsequent timesteps to explore underutilized semantic subspaces and enhancing temporal discriminability.

Unlike prior methods that stabilize or align timestep features through auxiliary consistency constraints, TRE explicitly diversifies temporal learning without architectural modifications or inference overhead. By progressively regulating the reuse of dominant patterns during training, TRE enables SNNs to acquire richer, more complementary, and temporally discriminative representations.

To summarize, our contributions are threefold:

- We provide a principled analysis revealing that temporally invariant supervision in conventional SNNs leads to temporal redundancy—where identical supervision across timesteps induces convergence of channel activations, causing gradient homogenization and diminishing temporal discriminability.
- We propose a novel Temporal Representation Enhancement (TRE) framework that redefines temporal learning as a learning-to-forget process. TRE dynamically suppresses high-contribution semantic features through a hybrid modulation strategy, encouraging each timestep to explore complementary representations.
- Extensive experiments on both static and dynamic benchmarks demonstrate that TRE consistently improves classification accuracy and temporal diversity across various SNN backbones, validating its effectiveness and generality.

2. Related Work

ANN-to-SNN Conversion. One prominent research direction for improving the representational ability of SNNs is the conversion of pre-trained Artificial Neural Networks (ANNs) into spiking counterparts [3, 4, 29, 44, 51]. By transferring the discriminative features learned through large-scale ANN training and advanced optimization, these conversion pipelines allow SNNs to inherit strong feature expressiveness with minimal additional supervision [1, 2, 17, 28, 39, 43, 53, 67]. Typical conversion techniques—including rate coding [3, 4, 29, 44, 51] and weight rescaling[3]—map continuous ANN activations into spike-based responses compatible with SNN firing dynamics [4, 9]. Despite their effectiveness, ANN-to-SNN conversion often relies on long simulation windows to approximate ANN behavior, leading to considerable computational overhead and increased inference latency. Motivated by these limitations, we focus on directly trained SNNs, which naturally exploit temporal dynamics while maintaining low latency and reduced energy consumption.

Temporal Learning and Supervision in SNNs. A second line of work examines how SNNs learn and exploit information over multiple timesteps. SNNs inherently generate a sequence of temporally evolving spike-based representations, yet most training paradigms employ uniform supervision across timesteps through surrogate-gradient backpropagation [47, 60]. Although effective for optimization, such temporally invariant supervision encourages timestep representations to converge toward similar subspaces, yielding highly correlated gradients and limited

temporal discriminability. To alleviate this issue, recent approaches leverage deeper architectures, multi-timestep calibration strategies, or consistency-enhanced temporal objectives [12, 13, 15]. However, these methods mainly aim to stabilize or align temporal features rather than explicitly diversify them. The persistent tendency toward temporal redundancy motivates our work: instead of enforcing uniform supervision, we promote timestep-specific representation exploration through a learning-to-forget paradigm that adaptively modulates decision-relevant patterns across timesteps.

Masking and Feature Modulation Mechanisms. Masking-based mechanisms have been used in CNNs to suppress dominant spatial responses and reveal alternative discriminative regions [36, 57, 66]. Related ideas appear in attention dropout [41], dropblock [19], and feature masking strategies [20]. In the context of SNNs, spatial erasing approaches such as temporal self-erasing have been shown to alleviate spatial redundancy [40, 59]. However, existing erasing or modulation mechanisms operate primarily in the spatial domain and do not target temporal redundancy. Our approach differs fundamentally by applying a temporally guided modulation strategy driven by class-specific contribution estimation, enabling timestep-wise enhancement through targeted suppression of high-contribution semantic patterns.

3. Method

In this section, we first analyze the temporal redundancy problem in conventional SNNs and discuss how it limits its feature diversity across timesteps. We then introduce the proposed *Temporal Representation Enhancement (TRE)* framework, which aims to promote complementary and discriminative temporal learning in SNNs, as illustrated in Fig. 2.

3.1. Temporal Redundancy in Conventional SNNs

During spiking neural computation, each timestep t generates a feature tensor $\mathbf{F}^t \in \mathbb{R}^{H\times W\times C}$, forming a temporal sequence of activations $\{\mathbf{F}^1, \ldots, \mathbf{F}^T\}$. Ideally, these feature maps should encode complementary information over time—early spikes capturing coarse semantics and later spikes refining fine-grained details. However, empirical observations reveal that SNNs often exhibit temporal inertia: the same high-response channels remain persistently active across timesteps, while weaker semantic pathways fail to emerge (as illustrated in Fig. 1).

To formalize this, we consider the temporal activation trajectory of channel c, measured by its average activation magnitude $a_c^t = \mathrm{mean}(|\mathbf{F}^t_{:,:,c}|)$. Without explicit temporal modulation, the evolution of a_c^t follows an exponential decay toward a stable equilibrium:

$$a_c^{t+1} = (1-\lambda)a_c^t + \lambda \bar{a}_c, \qquad 0 < \lambda < 1, \tag{1}$$

where $\bar{a}_c$ represents the steady-state response determined by the shared classifier and uniform supervision. This convergence behavior causes later timesteps to reproduce nearly identical activation distributions, thereby diminishing temporal discrimination.

From a representational standpoint, such repetitive dynamics imply that the mutual dependence between successive features increases:

$$I(\mathbf{f}^t; \mathbf{f}^{t-1}) \uparrow, \quad I(\mathbf{f}^t; y) - I(\mathbf{f}^{t-1}; y) \downarrow, \tag{2}$$

meaning that new timesteps contribute less novel information about the target class. This phenomenon, termed temporal redundancy, reflects a fundamental inefficiency—SNNs reuse past semantic responses instead of exploring complementary temporal cues.

To overcome this limitation, we introduce a mechanism that adaptively suppresses persistently dominant semantic channels, facilitating complementary feature learning across timesteps—the core principle of our proposed Temporal Representation Enhancement (TRE) framework.

3.2. Temporal Representation Enhancement

Building on the above analysis, we argue that timestep supervision should be differentiated rather than uniform, enabling SNNs to learn temporally diverse and complementary representations instead of repeatedly activating dominant patterns. From a cognitive neuroscience view [5], the brain refines perception through predictive coding—each cycle suppressing already-explained signals and emphasizing novel residual information.

While SNNs inherently process information across multiple timesteps, conventional training still drives them to focus on similar high-response channels throughout the sequence. To address this, we propose a Temporal Representation Enhancement (TRE) mechanism that dynamically suppresses over-dominant semantic channels and guides later timesteps to explore alternative discriminative cues, as shown in Fig. 2. Specifically, we denote the feature tensor at timestep t as $\mathbf{F}^t \in \mathbb{R}^{H\times W\times C}$, where C is the number of channels. Global average pooling (GAP) produces the channel descriptor:

$$\mathbf{f}^t = GAP(\mathbf{F}^t). \tag{3}$$

Temporal accumulation. To capture temporal evolution, we accumulate the averaged channel descriptors and logits from previous timesteps:

$$\hat{\mathbf{f}}^t = \frac{1}{t-1}\sum_{i=1}^{t-1}\mathbf{f}^i, \qquad \hat{\mathbf{S}}^t = \frac{1}{t-1}\sum_{i=1}^{t-1}\mathbf{S}^i, \tag{4}$$

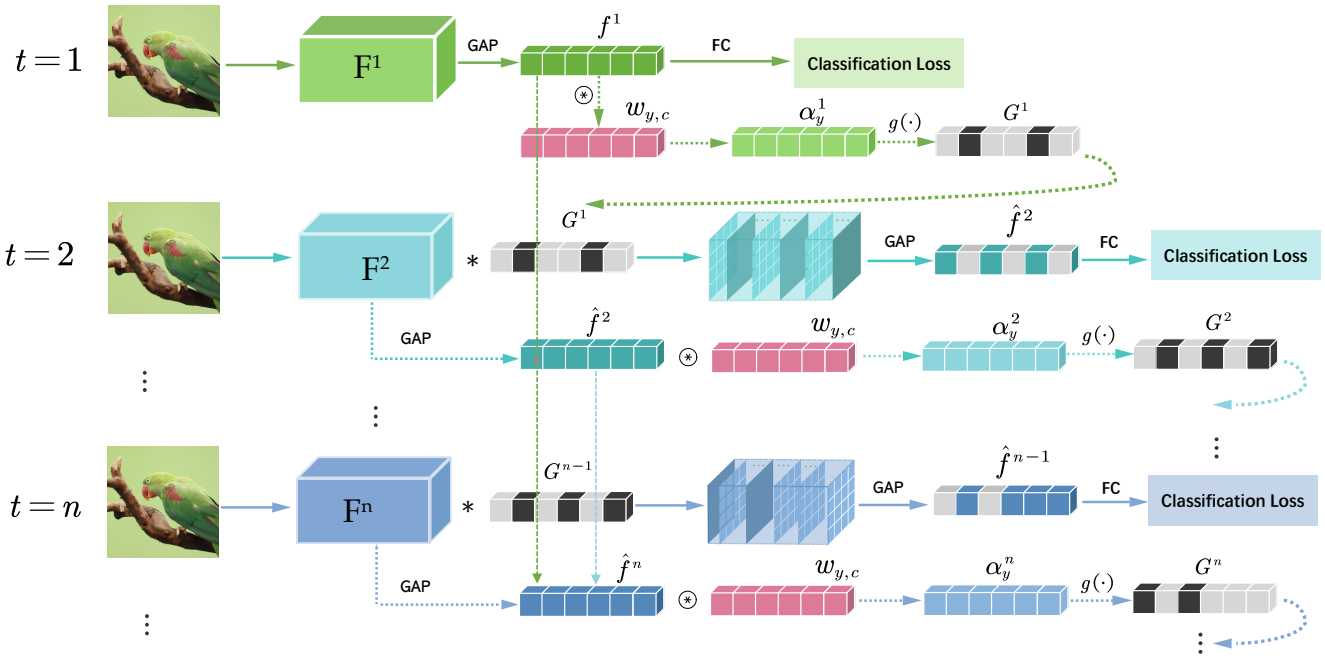


Figure 2. Overview of the proposed Temporal Representation Enhancement (TRE) framework. At each timestep t, TRE estimates class-specific channel contributions and generates a channel gating vector $\mathbf{G}^t$ to suppress highly contributing channels. This modulation guides later timesteps to focus on complementary semantic subspaces, enhancing temporal diversity and discriminability. During inference, the network operates without masking, adding no extra computational cost.

where $\mathbf{S}^i = \mathrm{FC}(\mathbf{f}^i)$ denotes the logits at timestep i and FC denotes the shared linear layer. These accumulated representations serve as a temporal prior for channel relevance estimation.

Class-specific contribution estimation. To quantify how much each channel contributes to the prediction of the ground-truth class y, we adopt a class-conditional attribution formulation. Each row vector $\mathbf{w}_y \in \mathbb{R}^C$ represents the classifier weights corresponding to class y, where each scalar $\mathbf{w}_{y,c}$ quantifies the evidence of the feature value in channel c for predicting that class. Accordingly, the relevance score for channel c at timestep t is defined as:

$$\alpha_y^t[c] = \frac{\left(\exp\left(\hat{\mathbf{f}}_c^t \cdot \mathbf{w}_{y,c}\right) - 1\right) \cdot \exp\left(\sum_{i=1, i\neq c}^{C} \hat{\mathbf{f}}_i^t \cdot \mathbf{w}_{y,i}\right)}{\sum_{k=1}^{K} \exp\left(\hat{\mathbf{S}}_k^t\right)}. \tag{5}$$

This relevance vector α_y^t measures the marginal change in confidence brought by each channel and acts as a timestep-aware semantic prior for identifying dominant patterns.

Adaptive Channel Gating. To suppress channels that dominate the current decision, we introduce an adaptive gating mechanism that selectively attenuates feature channels based on their class-specific contributions. The gating vector is defined as:

$$\mathbf{G}^t = g(\alpha_y^t), \tag{6}$$

where $g(\cdot)$ denotes a channel-wise gating function.

For stability, we combine a constant baseline threshold θ_0^c with a data-dependent threshold θ_a^c defined as:

$$\theta_a^c = \mu(\alpha_y^t) + \lambda_c\, \sigma(\alpha_y^t), \tag{7}$$

where $\mu(\alpha_y^t)$ and $\sigma(\alpha_y^t)$ denote the mean and standard deviation of the class-specific contribution scores, respectively, and λ_c is a channel-specific scaling factor that controls the sensitivity of the adaptive threshold. We then define a unified threshold:

$$\tau^c = \max(\theta_0^c, \theta_a^c). \tag{8}$$

Based on this threshold, the channel gating vector at timestep t is defined using an indicator function:

$$\mathbf{G}^t[c] = \mathbb{I}\left(\alpha_y^t[c] < \tau^c\right), \tag{9}$$

where $\mathbb{I}(\cdot)$ equals 1 when the condition is satisfied and 0 otherwise.

The gated feature tensor at timestep t is then obtained as:

$$\tilde{\mathbf{F}}^t = \mathbf{F}^t * \mathbf{G}^t, \tag{10}$$

where $*$ denotes broadcasted element-wise multiplication along the channel dimension. In this way, dominant channels with high contribution scores are suppressed, encouraging the model to forget dominant patterns at the current

timestep and explore complementary semantic cues across subsequent timesteps. The modulated features are then fed into the classifier to obtain $\tilde{\mathbf{p}}^t = \mathrm{softmax}(\mathrm{FC}(\mathrm{GAP}(\tilde{\mathbf{F}}^t)))$.

Training objective. For timestep $t>1$, we supervise the model using the modulated prediction:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(\mathbf{p}^1, y) + \sum_{t=2}^{T} \mathcal{L}_{\mathrm{CE}}(\tilde{\mathbf{p}}^t, y). \tag{11}$$

This objective encourages each timestep to extract new discriminative evidence rather than repeatedly relying on dominant channels.

4. Experiment

4.1. Datasets

To validate the generalization ability of our method, we evaluate it on both static image datasets and dynamic neuromorphic datasets. For static benchmarks, we use CIFAR-100 [33] and ImageNet [8], which represent standard large-scale visual recognition tasks. For dynamic benchmarks, we adopt DVS-CIFAR10 [35], an event-based neuromorphic dataset captured with a Dynamic Vision Sensor (DVS).

4.2. Experimental Settings

We evaluate the proposed method on three benchmark datasets: CIFAR-100, ImageNet, and DVS-CIFAR10. For CIFAR-100 and DVS-CIFAR10, we follow the standard direct-training protocol commonly adopted in prior SNN studies [6, 12]. We use LIF neurons and adopt ResNet-19 as the backbone. The firing threshold and membrane decay constant are set to 1 and 2, respectively. Models are trained using stochastic gradient descent with a momentum of 0.9 and an initial learning rate of 0.1, which is annealed to zero using a cosine decay schedule. For ImageNet, we follow the training configuration of PSN [15] and train all models for 320 epochs to ensure a fair comparison with state-of-the-art SNN approaches.

4.3. Comparisons with State-of-the-Art Methods

On CIFAR-100, our TRE achieves 81.27% accuracy with a spiking ResNet-19 using only 4 timesteps, surpassing SLTT, which uses 6 timesteps, by a margin of +6.60%, as summarized in Table 1. Compared with conversion-based approaches such as SlipReLU, which reaches 78.55% using 128 timesteps, TRE delivers superior accuracy with two orders of magnitude fewer timesteps, highlighting its efficiency in temporal representation learning. On ImageNet, TRE also delivers consistent gains across different backbones. Using ResNet-34, TRE achieves 71.60%, exceeding PSN by +1.04% under identical time-step settings. Compared with MPBN, TRE shows even larger improvements (71.60% vs. 64.71% on ResNet-34), demonstrating that temporal modulation provides strong benefits even for large-scale recognition tasks.

For dynamic neuromorphic data, TRE continues to exhibit superiority, as summarized in Table 2. On DVS-CIFAR10, TRE achieves 83.60% accuracy using ResNet-19 with 10 timesteps, outperforming the time-decoupled TET (83.00%) under the same training configuration. The consistent improvement across both frame-based and event-driven datasets shows that TRE effectively suppresses redundant dominant patterns and encourages more complementary temporal cues, enabling SNNs to exploit richer dynamics without increasing model complexity or inference cost.

4.4. Ablation Study

TRE effectiveness. To evaluate the effectiveness of the proposed Temporal Representation Enhancement (TRE) mechanism, we conduct ablation experiments on CIFAR-100 and DVS-CIFAR10 using ResNet-19 as the backbone. Table 3 compares TRE with the baseline training scheme and the TET supervision method under timestep settings of $T = 4$ and $T = 10$.

On CIFAR-100 (T=4), TRE improves the baseline accuracy from 77.75% to 81.27%, yielding a notable gain of +3.52 percentage points. Compared with TET, which enhances temporal optimization through decoupled timestep supervision and achieves 80.16%, TRE provides an additional +1.11 percentage point improvement. These results confirm that explicitly modulating dominant temporal patterns provides a more effective means of enriching multi-timestep representations than adjusting the supervision structure alone.

On DVS-CIFAR10 (T=10), a similar trend is observed. The baseline obtains 78.40%, while TET reaches 83.00%. TRE further boosts performance to 83.60%, surpassing the baseline by +5.20 percentage points and outperforming TET by +0.60. The larger improvement over the baseline on the event-based dataset highlights TRE's ability to encourage complementary temporal cues, which is particularly beneficial for neuromorphic data with inherently rich temporal structure.

Analysis of Gating Strategies. We further analyze different channel gating strategies to validate the effectiveness of the proposed design (as shown in Table 4). Specifically, we evaluate a rank-based top-k gating scheme (Top-1/2/4), where channels with the highest contribution scores are suppressed at each timestep.

Although the rank-based top-k gating strategy is effective, it consistently underperforms our adaptive threshold-based method. Moreover, performance degrades as more channels are suppressed, suggesting that a fixed top-k scheme tends to over-suppress informative features. In con-

Dataset	Method	Architecture	Time step	Train Method	Accuracy (%)
CIFAR-100	RTS [10]	ResNet-18	64	ANN-SNN conversion	69.27
	QCFS [3]	ResNet-18	4	ANN-SNN conversion	75.67
	SlipReLU [31]	ResNet-18	128	ANN-SNN conversion	78.55
	STBP-tdBN [68]	ResNet-19	6	Surrogate Gradient	71.12
	Sew ResNet [13]	ResNet-34	4	Surrogate Gradient	67.04
	TET [11]	ResNet-19	4	Surrogate Gradient	74.47
	InfLoR [22]	ResNet-20	5	Surrogate Gradient	71.19
	SLTT [45]	ResNet-18	6	Surrogate Gradient	74.67
	TTS [23]	ResNet-19	2	Surrogate Gradient	80.20
	GAC-SNN [49]	ResNet-18	4	Surrogate Gradient	79.83
	Ours	**ResNet-19**	**4**	**Surrogate Gradient**	**81.27**
ImageNet	Sew ResNet [13]	ResNet-18	4	Surrogate Gradient	63.18
		ResNet-34	4	Surrogate Gradient	67.04
	Real Spike [22]	ResNet-18	4	Surrogate Gradient	63.68
	MPBN [24]	ResNet-18	4	Surrogate Gradient	63.14
		ResNet-34	4	Surrogate Gradient	64.71
	PSN [15]	ResNet-18	4	Surrogate Gradient	67.63
		ResNet-34	4	Surrogate Gradient	70.54
	Ours	**ResNet-18**	**4**	**Surrogate Gradient**	**67.74**
		ResNet-34	**4**	**Surrogate Gradient**	**71.60**

Table 1. Comparison results with different methods on CIFAR100 and ImageNet.

Dataset	Method	Architecture	Time step	Train Method	Accuracy (%)
DVS-CIFAR10	Dspike [37]	ResNet-18	10	Surrogate Gradient	75.40
	InfLoR [25]	ResNet-19	10	Surrogate Gradient	75.50
	SLTT [45]	VGG-11	10	Surrogate Gradient	77.30
	STBP-tdBN [68]	ResNet-19	10	Surrogate Gradient	67.80
	RecDis-SNN [21]	ResNet-19	10	Surrogate Gradient	72.42
	Real Spike [22]	ResNet-19	10	Surrogate Gradient	72.85
	PLIF [14]	7-layer CNN	20	Surrogate Gradient	74.80
	MPBN [24]	ResNet-19	10	Surrogate Gradient	74.40
	MLF [16]	ResNet-19	10	Surrogate Gradient	70.36
	TEBN [12]	7-layer CNN	10	Surrogate Gradient	75.10
	IM-Loss [21]	ResNet-19	10	Surrogate Gradient	72.60
	LSG [38]	ResNet-19	10	Surrogate Gradient	77.90
	LM-H [26]	ResNet-19	10	Surrogate Gradient	79.10
	TET [11]	ResNet-19	10	Surrogate Gradient	83.00
	Ours	**ResNet-19**	**10**	**Surrogate Gradient**	**83.60**

Table 2. Comparison results with different methods on DVS-CIFAR10.

trast, our method adaptively identifies dominant channels via a data-dependent threshold, enabling the model to selectively forget dominant patterns while preserving complementary information.

To mitigate the potential distribution shift between training and inference, we normalize the gating operation during training to preserve the expected activation magnitude, similar to dropout.

As shown in Fig. 3, normalized gating introduces only a minor distribution shift in classification scores compared to inference, and significantly less than unnormalized gating. This prevents over-saturation of activations and stabilizes training, while maintaining consistency between training and inference.

Feature Redundancy Analysis. To further assess the impact of TRE on temporal representation diversity, we compare our method with the recent PSN [15] framework, which also adopts a temporal decoupling paradigm for SNNs. Specifically, we analyze the inter-timestep feature similarity between the PSN baseline (ResNet-34) and its TRE-augmented counterpart on the ImageNet dataset. As

Dataset	Model	T	Accuracy(%)
CIFAR-100	baseline	4	77.75
	TET [11]	4	80.16
	TRE	4	**81.27**
DVS-CIFAR10	baseline	10	78.40
	TET [11]	10	83.00
	TRE	10	**83.60**

Table 3. Performance of TRE compared with alternative supervision strategies on CIFAR-100 and DVS-CIFAR10, evaluated under timestep settings of T=4 and T=10.

Dataset	Model	Accuracy(%)
CIFAR-100	Top-1 gating	80.88
	Top-2 gating	80.82
	Top-4 gating	80.52
	TRE (ours)	**81.27**

Table 4. Performance comparison of different gating strategies on CIFAR-100 with timestep T=4.

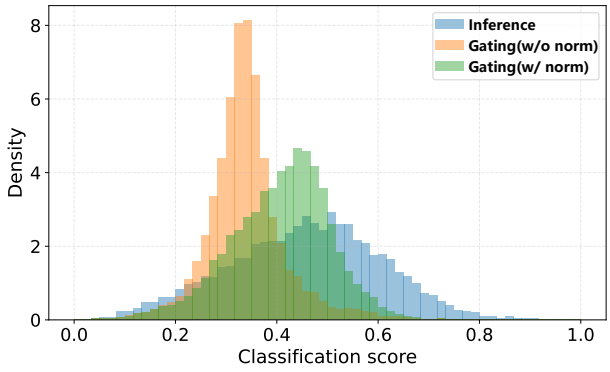


Figure 3. Classification score distributions for inference (no gating), unnormalized gating, and normalized gating.

shown in Table 5, the cosine similarity between features extracted from different timesteps is calculated to quantify temporal redundancy, where lower off-diagonal values indicate reduced correlation and enhanced diversity.

For the baseline PSN model using ResNet-34, the mean off-diagonal similarity is 0.95, revealing that features across timesteps are highly correlated and thus temporally redundant. After integrating TRE, the off-diagonal mean similarity decreases markedly to 0.85, indicating that the feature representations become more temporally diverse and discriminative. Notably, adjacent timestep pairs (e.g., T1–T2 and T3–T4) exhibit a pronounced similarity reduction, demonstrating that TRE effectively guides the model to explore complementary temporal cues rather than repeatedly activating dominant patterns. This analysis confirms that TRE successfully mitigates the redundancy inherent in PSN's temporal representations, yielding richer and more decorrelated temporal features.

The hyper-parameters. We further analyze the sensitivity of TRE to its two key hyper-parameters, the fixed threshold θ_0^c and the adaptive scaling factor λ_c, as summarized in Table 6. The fixed threshold θ_0^c determines the minimum relevance required for a channel to be suppressed: a larger θ_0^c corresponds to gating a greater number of channel features. Conversely, the parameter λ_c controls the strength of the dynamic adjustment, where a smaller λ_c leads to a lower adaptive threshold and thus a higher gating rate.

As shown in the table, the performance remains relatively stable across a reasonable range of values, demonstrating the robustness of TRE. Accuracy peaks when θ_0^c=0.05 and λ_c=0.1, achieving 81.27% on CIFAR-100. Extremely large θ_0^c or very small λ_c degrade performance, as excessive suppression removes informative channels, while overly relaxed thresholds reduce the regularization effect.

Training Overhead. We clarify that the claim of "no extra computational cost" refers to inference only. The proposed TRE introduces a small additional cost during training due to the channel gating operation. On CIFAR-100 with 320 training epochs, training with TRE takes 12:02:35, compared to 10:58:22 for the baseline, corresponding to an overhead of approximately 9.8%. In contrast, the inference time remains unchanged for both models (0:00:22 with batch size 128), since the gating mechanism is disabled at test time. These results show that TRE incurs only a modest one-time training overhead while introducing no additional inference cost.

4.5. Visualization

We qualitatively assess temporal diversity by comparing ResNet-34(PSN [15]) and ResNet-34+TRE on ImageNet. For each image and timestep $T \in \{1, 2, 3, 4\}$, we compute channel contribution scores at the final stage, transform them to nonnegative probabilities, and plot their 1D kernel density estimation(KDE). Fig. 4 overlays ResNet-34 (solid blue) and ResNet-34+TRE (dashed red), enabling a direct, per-timestep comparison.

Across all cases, ResNet-34+TRE exhibits (i) *broader, less peaked* distributions, indicating contribution mass distributed over more channels rather than concentrated on a few; (ii) *timestep-wise complementarity*, where curves remain dispersed from T=1 to T=4 instead of collapsing to an early-step profile; and (iii) *attenuation of dominant peaks*, consistent with TRE's learning-to-forget mechanism. These trends visually corroborate our quantitative results, showing that TRE decorrelates timestep features and enhances temporal discriminability without altering the backbone or adding inference overhead.

Timestep Pair	ResNet-34				ResNet-34 + TRE			
	T1	T2	T3	T4	T1	T2	T3	T4
T1	1.00	0.93	0.97	0.98	1.00	0.84	0.81	0.88
T2	0.93	1.00	0.91	0.94	0.84	1.00	0.85	0.89
T3	0.97	0.90	1.00	0.95	0.81	0.85	1.00	0.83
T4	0.98	0.94	0.95	1.00	0.88	0.89	0.83	1.00
Off-diagonal mean	**0.95**				**0.85**			

Table 5. Inter-timestep cosine similarity analysis of features from the final stage (Stage 4) for ResNet-34 and ResNet-34+TRE on ImageNet. Lower off-diagonal scores correspond to less temporal redundancy and greater feature diversity.

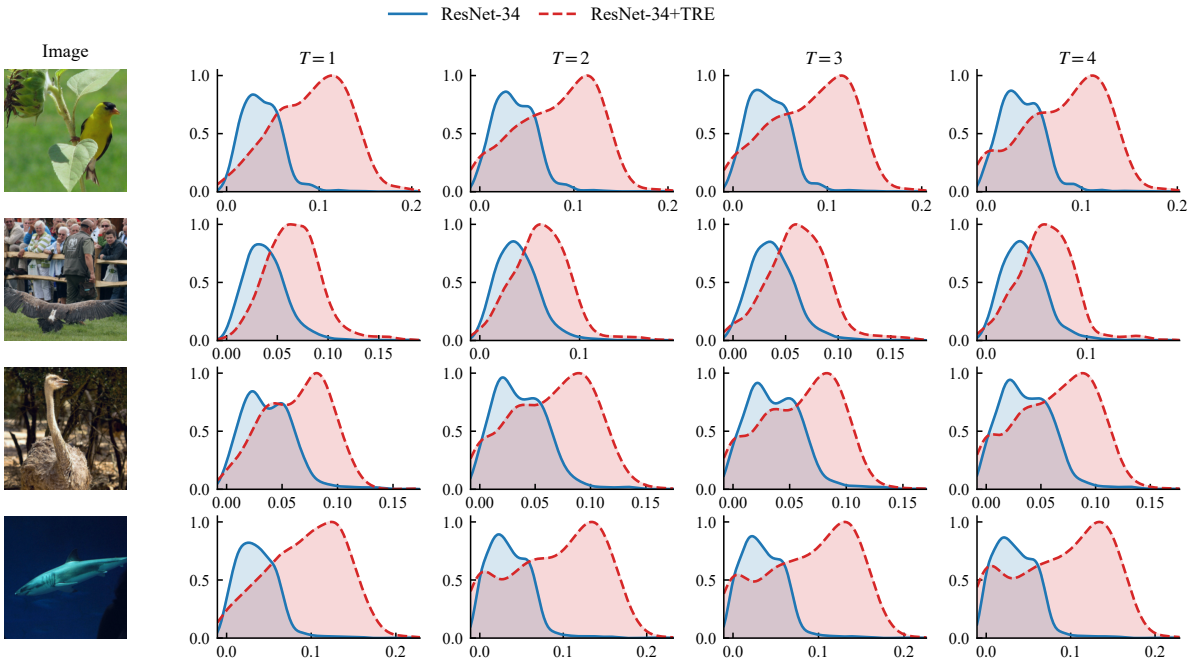


Figure 4. Kernel density estimation(KDE) visualization of channel contribution distributions for ResNet-34 vs. ResNet-34+TRE across timesteps (T=1 $\sim$ 4). Each row shows one image; dashed red curves (ResNet-34+TRE) are broader and less peaked than solid blue (ResNet-34), indicating reduced dominance and greater temporal diversity.

θ_0^c	λ_c	Accuracy(%)
0.04	0.0	80.99
	0.1	81.25
	0.2	81.03
0.05	0.0	80.76
	0.1	**81.27**
	0.2	81.27
0.06	0.0	81.01
	0.1	80.85
	0.2	80.67

Table 6. Performance comparison of adaptive channel gating methods on the CIFAR-100 dataset with time steps $T = 4$.

5. Conclusion

In this paper, we proposed *Temporal Representation Enhancement (TRE)*, a lightweight learning-to-forget framework that mitigates temporal redundancy in SNNs. By dynamically suppressing dominant channels and promoting complementary feature learning, TRE enhances temporal diversity without adding architectural or inference overhead. Comprehensive experiments on both static and dynamic datasets confirm consistent gains in accuracy and representational richness, demonstrating TRE's effectiveness as a general mechanism for improving temporal discriminability in neuromorphic models.

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